

SEDAR 15: South Atlantic Red Snapper, South Atlantic Greater Amberjack, and Florida Mutton Snapper Review

South Atlantic Red Snapper

Data Workshop

Life History Workgroup

- 1) Use new technology such as recent advances in genetics techniques (microsatellite multiplex panels; see Saillant and Gold (2006)) to reinvestigate the stock structure and estimate the effective population size of red snapper in the Gulf of Mexico and along the Atlantic coast.
- 2) Obtain better estimates of red snapper natural mortality and release mortality in commercial and recreational fisheries.
- 3) Investigate life history of larval/juvenile (age 0 and 1) red snapper, as little is known.
- 4) All future age assessments (any species) should include assessment of otolith edge type. Classification schemes for edge type and quality of the otolith/section have been developed by the MARMAP program (Table 2.1). These classifications are currently used by MARMAP and NMFS Beaufort.
- 5) Continue to conduct inter-lab comparison of age readings from test sets of otoliths in preparation for any future stock assessments.
- 6) Obtain adequate data for gutted to whole weight conversions a priori (before stock assessment data workshop).
- 7) Strategies for collection of ageing parts vary for estimations of age composition and von Bertalanffy growth parameters. Typically, small specimens from fishery independent sampling are needed to produce good estimates of von Bertalanffy parameters.

Commercial Workgroup

The following research recommendations were developed by the Working Group:

- Still need observer coverage for the snapper-grouper fishery
 - 5-10% allocated by strata within states
 - possible to use exemption to bring in everything with no sale
 - get maximum information from fish
- Expand TIP sampling to better cover all statistical strata
 - Predominantly from Florida and by H&L gear
 - In that sense, we have decent coverage for lengths
- Trade off with lengths versus ages, need for more ages (i.e., hard parts)

- Workshop to resolve historical commercial landings for a suite of snapper grouper species
 - Monroe County (SA-GoM division)
 - Species identification (not an issue with red snapper)

Recreational Workgroup

Six years of concurrent RDD and FHS effort estimates for east Florida need to be compared for adjusting effort estimates in for-hire mode for future assessments. This has been done in the Gulf for six years of concurrent data and resulted in significant changes to landings estimates for red snapper in the Gulf of Mexico assessment (SEDAR 7).

The PSE's for MRFSS estimates for reef-fish species continue to be high in the south Atlantic region, in spite of increased sample sizes implemented in recent years. The workgroup recommends evaluating recreational fishery survey data to study the relationship between sample size (both angler intercepts and effort interviews) and precision of annual catch estimates of reef-fish species at the sub-region and state levels to determine what sample sizes are needed to obtain minimum PSE levels of 20% or less.

Better geographic definition for estimated effort and catch are needed for red snapper in the south Atlantic. Red snapper are considered rare north of Cape Hatteras, NC. In Florida, red snapper are abundant in northeast Florida and less common in southeast Florida; however, private boat mode estimates are for the entire Florida east coast. The FHS stratifies east Florida into two subregions for better precision. Monroe County is a separate sub-region in the for-hire survey, but for private boat mode, MRFSS estimates effort and landings for the entire Gulf Coast of Florida, which included Monroe County. There is currently no way to separate Monroe County landings by Atlantic and Gulf waters in either the MRFSS or FHS. In addition to finer geographic scales, more detailed information on location of catch are needed from angler interviews. Currently, the MRFSS and FHS only delineate if fishing occurred in inland, state, or federal waters with no further detail on area fished or depth.

These issues come up repeatedly in data work shops and stock assessments for other species, and a finer scale stratification for data collection and sample distribution with more detailed area fished information should be pursued in efforts to refine and improve recreational data collections at the national level, which are currently underway.

Indices Workgroup

1. Develop a method to correct for red snapper that are misclassified or unclassified on a trip-by-trip basis.
2. Expand existing fishery independent sampling and/or development new fishery independent sampling of red snapper population so off the southeastern U.S. Two ideas discussed were the following:
 - Adding gears to MARMAP that are more effective at catching red snapper
 - Developing coast-wide sampling of larval and juvenile abundance

3. Examine how catchability has changed over time with increases in technology and potential changes in fishing practices. This is of particular importance when considering fishery dependent indices.
4. Investigate potential density-dependent changes in catchability.
5. Examine possible temporal changes in species assemblages. Such changes could influence how the Stephens and MacCall method is applied when determining effective effort.
6. Continue and expand the “Headboat at Sea Observer Survey”. This survey collects discard information, which would provide for a more accurate index of abundance.

Assessment Workshop

No research recommendations were reported for the Assessment Workshop

Review Workshop

The Review panel responded to Term of Reference 9:

9. Review the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment.

The RP supports the recommendations of data workshop. Of the recommendations provided in the report, the most critical priority for stock assessment is establishment of a fishery independent index. This could best be accomplished by adding gears to the MARMAP survey that are more effective at catching red snapper.

Other important recommendations are:

- Quantifying release mortality and length/age structure of discards, for instance by expanding the “Headboat at Sea Observer Survey.”
- Using consistent otolith ageing assumptions.
- Assessing the degree to which catchability has changed over time.
- Improving data collection protocols.

The recommendation to analyze stock structure using microsatellite genetic techniques, while good science, is probably less important to improving the current assessment.

The panel felt that the procedure for choosing the weights in the likelihood function might be improved and recommends that a more rigorous protocol be investigated to avoid criticism of subjectivity.

Bayesian methods should be considered for inference on uncertainty. These methods would allow priors on steepness, natural mortality, and other parameters to be chosen in order to quantify uncertainty in stock status and benchmarks. These additional procedures will require adequate time being afforded to assessment scientists to develop the appropriate tools.

In order to be able to measure an improvement in the stock, the next assessment would need to be conducted some years (5 perhaps) after any new management measures are introduced. This

implies an interval of about 6-7 years before the next assessment. If managers are particularly concerned about the status of the stock, then a shorter interval of 3 years might be considered to check whether any further deterioration has occurred, but this would not be a sufficiently long time interval to be able to detect the efficacy of management measures.

South Atlantic Greater Amberjack

Data Workshop

Life History Workgroup

- 1) Use new technology such as satellite pop-up archival tags and recent advances in genetics techniques to reinvestigate the mixing rate between greater amberjack in the Gulf of Mexico and those in the waters along the Atlantic coast of the southeastern U.S. Such research will also provide insight into post-release survivorship, migratory patterns, and spawning locations.
- 2) All future age assessments (any species) should include assessment of otolith edge type. Classification schemes for edge type and quality of the otolith/section have been developed by the MARMAP program at SCDNR (Table 2.3). These classifications are currently used by MARMAP and NMFS Beaufort.
- 3) Conduct inter-lab comparison of age readings from test sets of otoliths in preparation for any future stock assessments.
- 4) Obtain adequate data for gutted to whole weight conversions a priori (before stock assessment data workshop).
- 5) Obtain better estimates of greater amberjack natural mortality and release mortality in commercial and recreational fisheries.
- 6) Strategies for collection of ageing parts vary for estimations of age composition and von Bertalanffy (VB) growth parameters. Typically, small specimens from fishery independent sampling are needed to produce good estimates of VB parameters.
- 7) Investigate life history of larval/juvenile (age 0 and 1) greater amberjack, as little is known.

Commercial Workgroup

The following research recommendations were developed by the Working Group:

- Still need observer coverage for the snapper-grouper fishery
 - 5-10% allocated by strata within states
 - possible to use exemption to bring in everything with no sale
 - get maximum information from fish
- Expand TIP sampling to better cover all statistical strata
 - Predominantly from Florida and by handline gear
 - In that sense, we have decent coverage for lengths

- Trade off with lengths versus ages, need for more ages (i.e., hard parts)
- Workshop to resolve historical commercial landings for a suite of snapper-grouper species
 - Monroe County (SA-GoM division)
 - Species identification is a major issue with amberjack

Recreational Workgroup

Six years of concurrent RDD and FHS effort estimates for east Florida need to be compared for adjusting effort estimates in for-hire mode for future assessments. This has been done in the Gulf for six years of concurrent data and resulted in significant changes to landings estimates for red snapper in the Gulf of Mexico assessment (SEDAR 7).

The PSE's for MRFSS estimates for reef-fish species continue to be high in the south Atlantic region, in spite of increased sample sizes implemented in recent years. The workgroup recommends evaluating recreational fishery survey data to study the relationship between sample size (both angler intercepts and effort interviews) and precision of annual catch estimates of reef-fish species at the sub-region and state levels to determine what sample sizes are needed to obtain minimum PSE levels of 20% or less.

Better geographic definition for estimated effort and catch are needed for greater amberjack in the south Atlantic. There is currently no way to separate Monroe County landings by Atlantic and Gulf waters in either the MRFSS or FHS. Private boat estimates for Monroe County must be post-stratified from west Florida estimates. In addition to finer geographic scales, more detailed information on location of catch are needed from angler interviews. Currently, the MRFSS and FHS only delineate if fishing occurred in inland, state, or federal waters with no further detail on area fished or depth. These issues come up repeatedly in data work shops and stock assessments for other species, and a finer scale stratification for data collection and sample distribution with more detailed area fished information should be pursued in efforts to refine and improve recreational data collections at the national level, which are currently underway.

Indices Workgroup

1. Develop a method to correct for greater amberjack that are misclassified or unclassified on a trip-by-trip basis.
2. Expand existing fishery independent sampling and/or development new fishery independent sampling of greater amberjack population so off the southeastern U.S. Two ideas discussed were the following:
 - Adding gears to MARMAP that are more effective at catching greater amberjack
 - Developing coast-wide sampling of larval and juvenile abundance

3. Examine how catchability has changed over time with increases in technology and potential changes in fishing practices. This is of particular importance when considering fishery dependent indices.
4. Investigate potential density-dependent changes in catchability.
5. Examine possible temporal changes in species assemblages. Such changes could influence how the Stephens and MacCall method is applied when determining effective effort.
6. Continue and expand the “Headboat at Sea Observer Survey”. This survey collects discard information, which would provide for a more accurate index of abundance.

Assessment Workshop

No research recommendations were reported for the Assessment Workshop

Review Workshop

The Review panel responded to Term of Reference 9:

9. Review the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment.

The DW and AW made numerous recommendations regarding further research that might improve future assessments of mutton snapper. The review panel supports those recommendations, and in particular endorses the following:

- Collection of specimens for maturity analysis. After selection criteria had been applied to select an appropriate subset of potential samples, only 32 specimens were available to estimate the maturation schedule for the current assessment,
- Continued monitoring of discards in the commercial and recreational (headboat) fisheries to estimate magnitude and size frequency of discards is endorsed.
- Continuation of the various fishery independent surveys was recommended by the DW. The panel endorses this recommendation, but notes that the current surveys generally encompass only a portion of the habitats and regions of the mutton snapper stock, which may limit their utility for stock assessment. A fishery independent survey that encompasses the range of the stock would have greater value for stock assessment than a multitude of surveys that each are limited in geographic range.

The review panel noted the limited flexibility of the age-structured model (ASAP) used for the mutton snapper assessment and recommends that a more flexible age-based model be used in

future assessments. Particular functionality that was missing from the ASAP model includes: ability to model both asymptotic and dome-shaped selectivity; ability to fit length frequency data directly; ability to fit longer time series of data; and, ability to initialize the population assuming a constant historical exploitation rate. The RP encouraged the continued development of ASAP as it provides an accessible software platform that can be used by a wide range of users.

Florida Mutton Snapper (SEDAR 15A)

Data Workshop Research Recommendations

Life History Workgroup

The biology of *Lutjanus analis* during reproduction remains perhaps the greatest unknown in the life-history of this species. Despite its relatively large body size, exploited status, and gregarious nature during reproduction, the behaviors, location, and sources of individuals of spawning aggregations in Florida and the greater Caribbean remains elusive. Seasonal migration patterns are completely unknown and based on speculation. Primary habitats used by this species during various stages of its ontogeny are undefined. This information would reveal the dependence of the Florida population on various habitats and locations, e.g., a given spawning location; critical information since models have revealed that contributions to the Florida population of *L. analis* in the form of larvae from outside southern sources is minimal (Paris et al. 2005), and that the Florida population is biologically “on its own”. Because of the aforementioned difficulties and differences in staging criteria, we recommend further review of the maturity data from Tequesta and the Florida Keys, and Puerto Rico before accepting the size- and age- at-maturity values from the regressions reported here.

Commercial Workgroup

Increasing the dockside sampling of commercial catches, particularly for the longline and bandit rig fisheries will be important to monitoring the size of fish, areas and depths fished, and fishing effort for this species and other reef fish. The scarcity of otoliths in the earlier portions of the sampling time series restricts the amount of age information that could be used for assessments, and we suggest placing more emphasis on sampling otoliths for this and other reef species to aid future age-structured stock assessments. There is also a need for increasing the amount of discard information (either at-sea or from logbooks) and discard mortality data in modern stock assessments, including this species. Few discards of mutton snapper were actually noted in commercial fishermen’s logbooks, and perhaps the number of fish discarded by commercial fishermen is really low. However, the relatively low frequency of discard logbooks assigned to fishermen may have also been a factor in the low number of discard records provided. Mutton snapper tend to be caught in low numbers with other reef fish species, and relatively few commercial fishing trips actually appear to target this species.

An examination of the conversion factors used to convert landed weight to whole weight should be undertaken. A comparison of the regressions in Life History Section II (Table 2.12) for gutted weight and whole weight would appear to suggest a lower percentage difference between gutted weight and whole weight at comparable sizes, perhaps as low as 2-5% rather than the 11%

currently used for all snappers. However, at this time, there is not enough data to allow a direct comparison of gutted weight to whole weight and derive a suitable conversion factor and the differences suggested would be small and perhaps negligible for the stock assessment.

Ultimately, if allocation between the various sectors of the fishery for mutton snapper and other reef fish are contemplated, conversion factors may become more of an issue.

There were differences noted in the commercial fisheries landings data between the ALS system, the General Canvass data, and the FWC trip ticket data. These differences should be reconciled so that each system will provide comparable numbers where appropriate.

Recreational Workgroup

Biological sampling of recreational landings in Florida has been funded on the West Coast of Florida, including Monroe County, since 2000, but continues to remain unfunded on the East Coast of Florida. Improved biological data collections are essential for making use of the best stock assessment models currently available, and the Recreational Data Working Group recommends funding and implementation of biological data collections in the shore, private boat, and for-hire modes on the east coast of Florida. The Recreational Data Working Group recommends continued funding for discard data collection and improved data collections on depth and area fished in the Headboat At-Sea Survey in Florida. Data on discarded catch is particularly important for size and bag regulate species, such as mutton snapper. The Working Group also recommends better data collection for area and depth fished in the MRFSS. Depth and area fished are particularly important for calculating depth and area-dependent discard mortality rates for reef fish species, such as mutton snapper, that are found in progressively deeper habitats throughout their life history.

Indices Workgroup

GENERAL recommendations: Explore night fish data! No data taken at night by anyone!

Assessment Workshop Research Recommendations

Life History

The maturity analysis used in this assessment was based on only 32 fish. A study should be designed to collect mutton snapper for age and gonad samples at spawning sites during the spawning season. This would entail a multi-year study to identify the diurnal usage patterns at spawning sites during year and to collect gonad samples for histological examination. To maintain quality and ensure consistency among readers, a set of training histological slides should be developed.

Dependent Data Collections

It is essential that adequate numbers of aging structures be collected from all sectors of the fishery from all regions. A weakness of the assessment was the paucity of age samples in the 1980s and early 1990s.

Review Workshop Research Recommendations

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